RESEARCH CONTRIBUTION 23 OCTOBER 2026

# Thirteen integer variables for Diophantine ZFC consistency

Fixed-prime radical compression and a compatible positivity test

Prepared for Vladimir Reshetnikov and the ProveIt project.
AI-assisted research with ordinary mathematical proofs, independently reviewed reductions, literal polynomial circuits, fixed coefficients and exact checks.

**Abstract**

We give a fixed ordinary integer-coefficient polynomial in **thirteen unrestricted integer unknowns**, of exact total degree **1392**, with a **475-operation** arithmetic circuit. Composing the reductions proved here with the explicitly retained ProveIt native/compiler calibration proves in ordinary mathematics formalizable in ZFC that its integer solvability is equivalent to a finite syntactic ZFC refutation. Negating gives the requested equivalence between its lack of integer zeros and Con(ZFC).

The central mechanism uses distinct valuations at the prime 3 to combine several square conditions with fixed small weights. This replaces the size-dependent separation used by the inspected predecessor constructions. A normalized strong Pell witness makes a fourth root available for compression. A classical three-pentagonal representation gives a positivity radicand that is always 1 modulo 3, allowing a fifth root to join the same norm. The complete norm has degree 32 in its carrier; every radical and division disappears from the supplied polynomial circuit.

A companion uses **fourteen unrestricted integers**, exact degree **652**, and **230 operations**. We retain uncentered and partially centered control equations so that every degree-saving coordinate change is explicit. The six inherited numerical coefficient payloads total 615,883 uncompressed bytes and are reconstructed byte for byte.

This is a substantial new degree/variable tradeoff within the reviewed repository, not a proof of global simplicity or a new fewest-variable theorem. Ten-integer equations already exist at much higher degree. The new arithmetic proofs and finite verification are distinct from an end-to-end Lean/Rocq certificate or an exported derivation in ProveIt's encoded ZFC calculus; neither is claimed here.

Comparison snapshot: `cfbf118b4ad9b2c651fe96b571d9415261766328`.
All numbered packages 31–75 and the incoming research archives at that snapshot were included in the comparison.

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# 1 Result and comparison

## 1.1 The logical statement

Fix the recursive syntactic ZFC refutation predicate used by the retained ProveIt calibration. Its consistency sentence says that no finite refutation exists. The result is a fixed polynomial $P_{13} \in \mathbb{Z}[x_1, \ldots, x_{13}]$ with the following theorem:

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}^{13}\ P_{13}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

The proof first establishes the existence equivalence with a finite refutation and then negates it. It assumes neither consistency nor inconsistency. Integer quantification is the usual definable arithmetic quantification inside ZFC, including negative integers and zero.

The exact new arithmetic claim is independent of any assertion about ZFC: the new polynomial has a zero if and only if the retained calibrated source has one. The last source-to-refutation step uses named ordinary theorems from the pinned project. Their source texts, the exact parent polynomial, the literal machine and the coefficient recipe are retained in the archive. Section 8 distinguishes this mathematical composition from the remaining proof-assistant and object-language work.

## 1.2 Measured equations

Every operation in this table is one live binary addition, subtraction or multiplication. Squaring and multiplication by a fixed numeral count; numeral leaves and register reuse are free. Thus these are arithmetic-circuit counts, not bit-complexity estimates.

| Unrestricted-integer equation | Unknowns | Exact degree | Operations |
|---|---|---|---|
| New main equation, auxiliary root centered | 13 | **1392** | **475** |
| New uncentered control | 13 | 1408 | 473 |
| New companion, three roots centered | 14 | **652** | **230** |
| New integer-center control | 14 | 664 | 223 |
| New auxiliary-center control | 14 | 700 | 217 |
| New uncentered control | 14 | 748 | 215 |

An additional companion retains the ordinary strong root: it uses **15 unrestricted integers, degree 214, and 139 operations**. All seven equations have the same integer-zero existence status. Their different numbers of private positivity coordinates are stated explicitly; the named registers in their circuits are not additional unknowns.

## 1.3 What was already available

The following are selected previously delivered points from the actual comparison snapshot, including the incoming archives. They must not be replaced by the older comparison table in package 75.

| Source | Domain | Unknowns | Degree |
|---|---|---|---|
| Structural Round11 | $\mathbb{Z}$ | 37 | 4 |
| Structural Round11 | $\mathbb{Z}$ | 30 | 6 |
| Joint Guards | $\mathbb{Z}$ | 25 | 8 |
| Polynomial Shears | $\mathbb{Z}$ | 23 | 10 |
| Centered Pell Norms | $\mathbb{Z}$ | 22 | 14 |
| Polynomial Shears | $\mathbb{Z}$ | 21 | 16 |
| Pell Congruences | $\mathbb{Z}$ | 20 | 20 |
| Pell Congruences | $\mathbb{Z}$ | 19 | 28 |
| Pell Congruences | $\mathbb{Z}$ | 18 | 32 |
| Pell Congruences, immediate parent | $\mathbb{Z}$ | 17 | 88 |
| Package 67, named thirteen-integer form | $\mathbb{Z}$ | 13 | 10,461,744 |
| Polynomial Shears, few-variable alternative | $\mathbb{Z}$ | 10 | 1,081,125 |
| Radical Residue | $\mathbb{Z}_{>0}$ | 10 | 1152 |
| Polynomial Shears | $\mathbb{Z}_{>0}$ | 11 | 211 |
| Saturation and Projection, lower-degree nine | $\mathbb{Z}_{>0}$ | 9 | 1,103,950 |

The new thirteen-integer form lowers the degree of the named thirteen-integer predecessor by a factor exceeding 7500. The existing ten-integer forms remain better in variable count; the existing seventeen-integer form remains better in degree. Domain restrictions are material: the ten-positive degree-1152 equation is not a ten-unrestricted-integer equation.

The new mechanism was not found in the inspected repository or incoming texts. We make no worldwide priority claim. The pentagonal theorem used below is classical and is expressly attributed. The inherited native encoding and K215 calibration are retained; a completely different ZFC verifier has not been compiled in this contribution.

# 2 The complete thirteen-variable equation

## 2.1 Variables and fixed coefficients

The independent variables are exactly

$$j, C, z, h, r, v, w, T, c, s, u_1, u_2, \xi \in \mathbb{Z}. \tag{2}$$

All other symbols below are finite polynomial abbreviations in these thirteen integers. The six fixed coefficients are

$$(\beta, K_0, M_C, M_F, A_0, B_0), \qquad \beta = 2^{1064921} - 1, \quad A_0 = 2^{1113}, \quad B_0 = 2^{666104}.$$

The middle three integers belong to the complete source34 *mixed-parity-unrounded-1064921* tuple. Their exact byte values and finite constructive definition are included in `coefficients/`. $M_F$ is already shifted by $\beta$. These are fixed numerals, not parameters quantified along with the unknowns. No coefficient depends on waiting for a halting computation. The literal JSON names $u_1, u_2$ as `p1,p2`, respectively, and names $\xi$ as `xi`; these are only typographical renamings.

## 2.2 Native and Pell abbreviations

Define, in this order,

$$\begin{aligned}
&J = 6j + 1, && q = \beta J + 1, && W = 4(3w + 1),\\
&F = (K_0 + W)C - z(q - 1), && X = qW, && Y = (2v + 1)q^3,\\
&R = \big(q(q - F) - C\big)(q^2 - 1) + (M_C + qM_F)J, && && \\
&E = XY, && a = E + Y, && H = 4a + 3,\\
&\Delta = a^2 + H = (a + 2)^2 - 1, && k = R + 1 + hE, && \\
&\eta = c - kY, && \zeta = k - \eta, && \ell = EkY.
\end{aligned} \tag{3}$$

The normalized strong and auxiliary expressions are

$$\begin{aligned} I_0 &= (2r+1)c^2, & D_f &= 1+\Delta I_0^2,\\ Q &= \Delta^2 I_0^2, & \delta &= Q-1,\\ V &= D_f(cT-R)-c, & D_0 &= X+ac,\\ A_\tau &= \ell(\ell+k)+1, & D_m &= 1+\Delta c^2. \end{aligned} \tag{4}$$

The two sign quantities and their joint positivity test are

$$\begin{aligned} G &= A_0FB_0C-(q-A_0F-B_0C)^2,\\ Z &= \eta\zeta,\\ N &= GZ-(s^2-G-Z+1)^2,\\ S &= 24N-21-(6u_1-1)^2-(6u_2-1)^2. \end{aligned} \tag{5}$$

Finally set

$$\begin{aligned} p &= D_0+H(\xi+27V),\\ A &= 9H^2A_\tau, & B &= 81H^2D_f, & D &= D_m,\\ L &= 6561H^2S, & C_* &= 729H^2(QV^2-1). \end{aligned} \tag{6}$$

The letter $L$ here is only a radicand abbreviation; it is unrelated to the native block length in the coefficient generator.

## 2.3 A finite division-free norm definition

For $m \geq 0$ define the universal polynomial

$$\mathcal{F}_m(x;d_1,\dots,d_m) = \prod_{\varepsilon\in\{-1,1\}^m}\left(x-\sum_{i=1}^{m}\varepsilon_i\sqrt{d_i}\right), \qquad \mathcal{F}_0(x)=x. \tag{7}$$

The product is invariant under changing each radical's sign, so every radical occurs only to an even power. Therefore $\mathcal{F}_m \in \mathbb{Z}[x,d_1,\dots,d_m]$. Its degree in $x$ is $2^m$. This is a definition of an ordinary polynomial, not an instruction to use radicals in the final equation.

For a completely division-free finite prescription, expand

$$\mathcal{F}_4(p+t;A,B,D,L) = \sum_{j=0}^{16} b_j t^j.$$

Every $b_j$ is an integer polynomial in $p,A,B,D,L$. Define

$$\begin{aligned} U &= \sum_{j=0}^{8} b_{2j}C_*^j\delta^{8-j},\\ V_* &= \sum_{j=0}^{7} b_{2j+1}C_*^j\delta^{7-j}. \end{aligned} \tag{8}$$

The promised equation is

$$\boxed{P_{13} = U^2-\delta C_*V_*^2 = 0.} \tag{9}$$

Only finite sums, integer constants, additions, subtractions and fixed nonnegative powers occur. No new unknowns are hidden in $U,V_*$ or in the $b_j$. The supplied 475-operation graph expands this prescription into binary arithmetic instructions.

For proof, where $\delta \neq 0$, the defining identity is

$$P_{13} = \delta^{16}\mathcal{F}_5(p; A, B, D, L, C_*/\delta). \tag{10}$$

The uncentered control replaces $p$ by $D_0 + H\xi$. This changes only the coordinate $\xi$: the map $\xi \mapsto \xi + 27V$ is an integral polynomial bijection, since $V$ does not contain $\xi$.

**Theorem 2.1.** *For the six fixed numerals specified above, $P_{13}$ has an integer zero if and only if the retained source34 native construction has an accepting certificate. By its retained ordinary calibration, this is equivalent to a finite syntactic ZFC refutation. The exact degree of $P_{13}$ is 1392 and the supplied circuit has 475 live binary operations.*

# 3 Fixed-prime separation of square roots

The following elementary lemma is the central new tool in this construction. It is deliberately stated for signed radicands. Before the guard is recovered, two radicands could be negative, so an argument that merely counts imaginary summands would be insufficient.

**Lemma 3.1** (Valuation-separated rational roots)**.** *Let $d_1, \dots, d_m \in \mathbb{Q}^\times$. Suppose their valuations at a prime $p_0$ are distinct even integers. If, for some choices of square roots in $\mathbb{C}$, their signed sum is rational, then every $d_i$ is a square in $\mathbb{Q}$.*

*Proof.* Group the radicands by their classes in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$. For a class represented by $d$, write $\sqrt{d_i} = a_i\sqrt{d}$ with $a_i \in \mathbb{Q}^\times$, absorbing signs into $a_i$. Since

$$2v_{p_0}(a_i) = v_{p_0}(d_i) - v_{p_0}(d),$$

the $a_i$ in a class have pairwise distinct valuations. Their signed sum cannot vanish: after dividing by its unique term of smallest valuation, one obtains a rational unit plus terms divisible by $p_0$.

Distinct square classes yield linearly independent square roots over $\mathbb{Q}$. One way to see this is to pass to the finite multiquadratic extension generated by the roots and average a relation against its distinct quadratic characters. Thus a rational total sum requires the coefficient of each nontrivial square class to vanish. We just proved that no nonempty class can have zero coefficient. Only the trivial square class remains. The argument includes negative square classes and uses no order on the radicands. □

**Corollary 3.2.** *If the $d_i$ satisfy the lemma, then $\mathcal{F}_m(x; d_1, \dots, d_m) = 0$ with rational $x$ implies that all $d_i$ are rational squares.*

*Proof.* Over a splitting field the norm is the displayed finite product. A zero factor gives the rational signed sum needed by the lemma. □

The weights need not make any root large in the usual absolute value. They need only give distinct prime valuations inside every possible square class. This is what permits small fixed multipliers $3, 9, 27, 81$ even when the Pell roots vary over arbitrarily large sizes.

## 3.1 Why the five radicands are separated

The literal definitions give, for every integer assignment,

$$q \equiv 2 \pmod 3, \quad W \equiv 1 \pmod 3, \quad X \equiv 2 \pmod 3, \quad a \equiv \Delta \equiv H \equiv 0 \pmod 3.$$

Put $b = XY^2$. Modulo 3, $b$ is either 0 or 2, so

$$A_\tau = 1 + b(b+1)k^2 \equiv 1 \pmod 3.$$

Also $D_f, D_m, S \equiv 1 \pmod 3$, while $Q \equiv 0 \pmod 3$ and

$$B_y := \frac{QV^2 - 1}{Q - 1} \equiv 1 \pmod 3.$$

In particular $\delta = Q - 1 \neq 0$ *before* any positivity condition has been recovered. This justifies denominator clearing without a circular appeal to the guard.

Let $e = v_3(H) \geq 1$. As $H = 4a + 3$ cannot vanish for integral $a$, the five weighted squared roots in (10) have valuations

$$2e + 2, \qquad 2e + 4, \qquad 0, \qquad 2e + 8, \qquad 2e + 6.$$

They are pairwise distinct even integers. Hence a zero of $P_{13}$ makes

$$A_\tau, \quad D_f, \quad D_m, \quad S, \quad B_y$$

all rational squares. The first four are integers and therefore have integer square roots. The auxiliary rational square requires the descent in Section 5.2.

# 4 A positivity test compatible with the same prime

**Lemma 4.1** (Joint positivity)**.** *For integers $x, y$,*

$$x > 0 \text{ and } y > 0 \quad \Longleftrightarrow \quad \exists s \in \mathbb{Z} : \; xy - (s^2 - x - y + 1)^2 > 0.$$

*Proof.* Positivity of the expression gives $xy > 0$. If both arguments were negative, then $s^2 - x - y + 1 \geq |x| + |y| + 1 > \sqrt{|xy|}$, a contradiction. For the converse put $n = x + y - 1 \geq 1$. Then $n \leq xy$. Choose a square nearest to $n$. Its squared error is less than $n$: between $a^2$ and $(a + 1)^2$, with integral $n$, the nearest error is at most $a$ and $n > a^2$, unless $n$ is itself a square, when the error is zero. Thus the required expression is positive. □

**Lemma 4.2** (A 3-adic-unit positivity radicand)**.** *For any integer $N$,*

$$N > 0 \quad \Longleftrightarrow \quad \exists u_1, u_2 \in \mathbb{Z} : \; 24N - 21 - (6u_1 - 1)^2 - (6u_2 - 1)^2 \text{ is a square in } \mathbb{Q}.$$

*The displayed radicand is* 1 *modulo* 3 *for every assignment.*

*Proof.* An integer that is a rational square is an integer square. If the radicand equals $t^2$, then $24N - 21 = (6u_1 - 1)^2 + (6u_2 - 1)^2 + t^2 \geq 2$, so $N \geq 1$. The residue assertion follows immediately.

For completeness put $M = 24N - 21 = 24(N - 1) + 3$. Remove the largest power $9^e$ dividing $M$, obtaining $M_0$. We have $M_0 > 0$, $M_0 \equiv 3$ modulo 8, and $9 \nmid M_0$. Legendre's three-square theorem supplies $M_0 = x^2 + y^2 + z^2$. All three coordinates are odd and they are not all divisible by 3.

Use the integer matrix

$$L_0 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{pmatrix}, \qquad L_0^{\mathsf{T}} L_0 = 9I_3.$$

For a vector nonzero modulo 3, independent sign choices make $x - y - z \neq 0$ modulo 3. Otherwise subtracting two choices that differ in one sign would force all three coordinates to vanish. Modulo 3 the three rows of $L_0$ give $x - y - z$, its negative, and itself. Thus, after choosing signs, applying $L_0$ makes all three coordinates nonzero modulo 3, preserves their oddness, and multiplies the squared norm by 9. Repeat $e$ times. If $e = 0$, the sum of the three squares is 0 modulo 3, and the fact that the coordinates are not all divisible by 3 already makes all three nonzero modulo 3. Finally choose signs so that each coordinate is $-1$ modulo 6. We have

$$M = (6u_1 - 1)^2 + (6u_2 - 1)^2 + (6u_3 - 1)^2,$$

as required. □

This is the classical theorem that every nonnegative integer is a sum of three generalized pentagonal numbers, using $24\,x(3x - 1)/2 + 1 = (6x - 1)^2$. Guy's observation and the norm-9 lifting identity are discussed and proved in Sun [4]; see also Oh [5]. Neither that theorem nor the lifting identity is claimed as new. Its uniformly nonzero residue is the feature exploited in the present five-root merger.

# 5 Soundness: every new zero restores a source zero

Assume $P_{13} = 0$ at an arbitrary signed integer assignment. The preceding valuation argument supplies signed rational roots $t, f, d, t_+, y$ satisfying

$$t^2 = A_\tau, \quad f^2 = D_f, \quad d^2 = D_m, \quad t_+^2 = S, \quad y^2 = B_y, \tag{11}$$

$$D_0 + H(\xi + 27V) = 3Ht + 9Hf + d + 81Ht_+ + 27Hy. \tag{12}$$

The signs in this equation are absorbed into the roots. Since their first four squares are integers, $t, f, d, t_+$ are integers.

## 5.1 Recover positivity and exclude every singular branch

By Lemma 4.2, $N > 0$; by Lemma 4.1, $G > 0$ and $Z = \eta\zeta > 0$. In particular $k = \eta + \zeta$ is nonzero, its absolute value is at least 2, and $0 < \eta/k < 1$.

The literal affine forms exclude $q = 0$ and $W = 0$. Moreover $q$ is even, $W$ is a nonzero multiple of 4, and $Y = (2v + 1)q^3$ is a nonzero multiple of 8. Hence $|X| \geq 8$ and $|Y| \geq 8$. From $c/k = Y + \eta/k$ we obtain $|c| > 2$. Also $a = (X + 1)Y$ is a multiple of 24, so $\Delta = (a + 2)^2 - 1 \geq 3$ is odd. Since $2r + 1 \neq 0$,

$$Q = \Delta^2(2r + 1)^2c^4 > 1.$$

All these facts precede the rational-Pell integrality step.

## 5.2 Rational auxiliary roots are integral

**Lemma 5.1** (Adjacent-coefficient Pell descent)**.** *If $Q > 1$ and $V$ are integers and $y \in \mathbb{Q}$ satisfies*

$$QV^2 - (Q - 1)y^2 = 1,$$

*then $y \in \mathbb{Z}$.*

*Proof.* Change signs to take $V, y > 0$. The equation excludes $V = 0$. Write $y = b/d$ in lowest terms with $d > 0$. Multiplying by $d^2$ shows $d^2 \mid Q - 1$. For $V = 1$ the equation gives $y = 1$. If $V > 1$, define

$$V' = (2Q - 1)V - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2QV.$$

Then $V'$ is an integer, the same norm equation holds, and the reduced denominator of $y'$ is still $d$, because $2Q - 1 \equiv 1 \pmod{d}$. The original equation gives $y > V$ and $QV > (Q - 1)y$, whence $-V < V' < V$. The norm excludes $V' = 0$. Induction on $|V|$ reaches $|V| = 1$ while preserving the denominator, proving $d = 1$. □

This argument is inherited from the rational Pell merger in package 69; it is reproduced here because its exact hypotheses matter. Apply it to (11); thus $y$ is an integer as well.

## 5.3 Explicit restoration maps

Equation (12) now implies $H \mid d - D_0$. Define

$$\begin{aligned} \gamma_{\text{old}} &= \xi + 27V - 3t - 9f - 81t_+ - 27y, \\ I_{\text{old}} &= \Delta I_0, & r_{\text{old}} &= \frac{\Delta(2r + 1) - 1}{2}, \\ T_{\text{old}} &= fT, & y_{\text{gap}} &= y - V, \\ \tau_{\text{half}} &= 2t - (2\ell + k), & J_{3,\text{old}} &= 2j. \end{aligned} \tag{13}$$

The only displayed division has an even numerator because $\Delta$ is odd. Every restored coordinate is integral.

The first Pell residual is restored by the identity

$$\tau_{\mathrm{half}}^2 + 2(2\ell + k)\tau_{\mathrm{half}} + k^2 - 4 = 4\big(t^2 - \ell(\ell + k) - 1\big) = 0.$$

The main root is $d = X + ac + H\gamma_{\mathrm{old}}$ and satisfies $d^2 - \Delta c^2 = 1$. The normalized strong square gives

$$I_{\mathrm{old}}^2 = \Delta(f^2 - 1), \qquad I_{\mathrm{old}} = (2r_{\mathrm{old}} + 1)c^2.$$

The CRT auxiliary expression becomes exactly

$$fcT_{\mathrm{old}} - f^2R - c = f^2(cT - R) - c = V.$$

The auxiliary norm gives the parent's gap residual because

$$V^2 + (1 - I_{\mathrm{old}}^2)y_{\mathrm{gap}}(2V + y_{\mathrm{gap}}) - 1 = I_{\mathrm{old}}^2V^2 - (I_{\mathrm{old}}^2 - 1)y^2 - 1 = 0.$$

Finally $N > 0$ admits a representation

$$N = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

For example apply the three-square theorem to $4N - 3 \equiv 1 \pmod 4$, put its unique odd coordinate last, and divide the other coordinates by 2. This restores the exact parent's final positivity row. All other parent definitions are literal polynomial identities. Thus every new zero restores a zero of the seventeen-integer CRT parent, whose source-to-refutation calibration is retained.

# 6 Completeness: an explicit normalized canonical lift

Soundness was a restoration of arbitrary signed zeros. Completeness uses the canonical witnesses supplied by the inherited native theorem. These are different logical operations: the new restriction need not preserve every old zero on the same coordinates.

## 6.1 The inherited canonical interface

A finite refutation supplies a source34 accepting tuple with

$$q = 2^t, \quad t \text{ odd}, \quad R \geq 3, \quad R \equiv 3 \pmod 4, \quad X = 2^R,$$

and $Y = (2v + 1)q^3 > 0$, together with $G > 0$, $Z > 0$, the first Pell root and the main root. Write $A_m = a + 2$ and

$$d = \chi_{A_m}(R), \qquad c = \psi_{A_m}(R), \quad \chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

The determinant guard's support estimates are uniform over this same canonical family. The main-root congruence gives an integral $\gamma_{\mathrm{old}}$.

The canonical native repunit $J$ is odd and 1 modulo 3, hence 1 modulo 6. Thus $J = 6j + 1$ loses no canonical witness. The canonical $W/4 = 2^{R-t-2}$ has even nonnegative exponent and is 1 modulo 3, so $W = 4(3w + 1)$ remains valid. These facts are explicitly recorded in the retained source70 orientation proof. Since $q$ is even, $a$ and $A_m$ are even, while $c$ is positive and odd.

## 6.2 Construct the odd normalized strong witness

Set

$$m = Rc, \quad I_0 = \psi_{A_m}(m), \quad f = \chi_{A_m}(m).$$

We claim $I_0/c^2$ is a positive odd integer. Pell composition gives

$$I_0 = c\,\psi_d(c), \qquad d^2 \equiv 1 \pmod c.$$

The recurrence for $\psi$ proves $\psi_d(n) \equiv nd^{n-1} \pmod c$, so $c \mid \psi_d(c)$. Therefore $c^2 \mid I_0$. Modulo 2, the recurrence at even $A_m$ gives $\psi_{A_m}(n) \equiv n$. Both $m$ and $c$ are odd, so the quotient $i = I_0/c^2$ is odd. Put $r = (i - 1)/2$. The Pell identity now gives $f^2 = 1 + \Delta I_0^2 = D_f$.

This is an explicit choice of a new strong witness. It does not assume that the old one happened to have the normalization needed for compression.

### 6.3 Construct the squared-modulus auxiliary witness

Put $M = \Delta I_0$ and choose

$$V^\dagger = \frac{\chi_M(R)}{M}, \qquad y = \psi_M(R).$$

For odd $R = 2b + 1$, the quotient is an integer polynomial $Q_b(M^2)$. The standard odd-index identities are

$$Q_b(0) = (-1)^b R, \qquad Q_b(1 - A_m^2) = (-1)^b \psi_{A_m}(R).$$

Since $R \equiv 3 \pmod 4$, $b$ is odd. Also $M^2 = \Delta(f^2 - 1) \equiv 1 - A_m^2 \pmod{f^2}$ and $c \mid M$. Consequently

$$V^\dagger \equiv -c \pmod{f^2}, \qquad V^\dagger \equiv -R \pmod c.$$

Because $f^2 \equiv 1 \pmod c$, the positive number

$$T = \frac{V^\dagger + c + Rf^2}{cf^2}$$

is an integer. It makes $V = D_f(cT - R) - c = V^\dagger$. The Pell identity at $M$ gives $QV^2 - (Q-1)y^2 = 1$. This squared-modulus argument is also supplied in source61, Section 5 of its signed-index proof. It is independent of the native coefficient tuple.

### 6.4 Choose the positivity and carrier coordinates

The pair $G, Z > 0$ permits an integer $s$ with $N > 0$ by Lemma 4.1. Lemma 4.2 then supplies $u_1, u_2, t_+$ with $t_+^2 = S$. Retain the canonical first Pell root $t$. Set

$$\xi = \gamma_{\text{old}} + 3t + 9f + 27y + 81t_+ - 27V.$$

This is an integer and makes (12) hold. One factor of the five-root norm is zero, so $P_{13} = 0$. Every new coordinate has been chosen explicitly or by a proved integer representation theorem. This completes both directions of Theorem 2.1.

## 7 Exact degree and the fourteen-variable companion

### 7.1 Uncentered degree

Assign degree one to each independent integer coordinate and degree zero to the six numerical ports. Direct substitution gives

| Register | $F$ | $R$ | $Y$ | $E, a, H$ | $k$ | $\ell$ | $I_0$ | $D_f$ | $Q$ | $V$ | $Z$ | $S$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Degree | 2 | 5 | 4 | 6 | 7 | 17 | 3 | 18 | 30 | 23 | 22 | 44 |

In the rational norm, the five weighted roots have degrees at infinity $23, 15, 7, 28, 29$ respectively. The last is uniquely largest. Thus the uncentered norm has degree $32 \cdot 29 + 16 \cdot 30 = 1408$ after clearing the auxiliary denominator. Equivalently, the highest term is $C_*^{16}$, with $\deg C_* = 88$.

### 7.2 The degree-1392 shear

Centering the auxiliary root is the integral coordinate change $\xi \mapsto \xi + 27V$. Since

$$B_y = V^2 + \frac{V^2 - 1}{Q - 1},$$

one formal branch has $\sqrt{B_y} = V + O(t^{-7})$ after all variables are scaled by $t$. Its weighted correction $27H(\sqrt{B_y} - V)$ has degree at most $-1$. Of the 32 factors, 16 retain the leading auxiliary term of degree 29, and 16 have it canceled. In the latter factors the positivity root, of degree 28, is uniquely dominant. Therefore

$$\deg P_{13} \le 16 \cdot 29 + 16 \cdot 28 + 16 \cdot 30 = 1392.$$

This argument may be read in a formal Laurent-series extension of the rational function field in the homogeneous input coordinates. It is a degree calculation, not an approximation used in the soundness proof.

For an explicit nonvanishing leading homogeneous form, let $H_0, V_0, Q_0, S_0$ denote the leading homogeneous parts. Then the leading part of $P_{13}$ is

$$Q_0^{16}(54H_0V_0)^{16}(6561H_0^2S_0)^8. \tag{14}$$

Each factor is nonzero. For example, if $k_0$ and $Y_0$ denote leading parts, then $Z_0 = -k_0^2Y_0^2$ and $S_0 = -24Z_0^2 \neq 0$. Similarly $V_0 = -(D_f)_0R_0 \neq 0$. This proves equality in the degree bound after the actual nonzero coefficient specialization. Independent exact univariate polynomial calculations modulo a prime check the stated degree and a nonzero leading coefficient of the delivered circuit.

## 7.3 Fourteen integers and degree 652

Keep three positivity coordinates $p_1, p_2, p_3$ in place of $u_1, u_2$, and omit the fifth radicand. There are now fourteen unknowns. Define

$$G_0 = N - \mathrm{Pos}(p_1, p_2, p_3), \qquad \mathrm{Pos}(p_1, p_2, p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).$$

Use the carrier

$$p_{14} = 2D_0 + H\{\xi + 6\ell + 3k + 18(a+2)I_0 + 54V\}.$$

Let

$$N_{14} = \delta^8 \mathcal{F}_4\left(p_{14}; 36H^2A_\tau, 324H^2D_f, 4D_m, \frac{2916H^2(QV^2-1)}{\delta}\right).$$

The same Taylor prescription as (8), now with the three-root norm of degree eight, specifies $N_{14}$ as an integer polynomial. The companion is

$$\boxed{(1 + 4G_0)(1 + 4N_{14}) - 1 = 0.} \tag{15}$$

Both factors are integers congruent to 1 modulo 4. Their product can equal 1 only when both equal 1. Thus this equation is exactly $G_0 = N_{14} = 0$.

The four roots have distinct valuations, as before; multiplying all weights by 2 does not change that. The carrier recovers $H \mid 2(d - D_0)$, hence $H \mid d - D_0$ because $H$ is odd. Conversely an old source tuple supplies an integer $\xi$ by the displayed carrier identity. No new parity restriction on the free carrier is needed.

Centering the auxiliary, first, and normalized strong roots gives a norm of exact degree 608. Its 16 branches split into groups of sizes $8, 4, 2, 2$, with respective degrees $29, 23, 15, 7$, and denominator contribution $8 \cdot 30$:

$$8 \cdot 29 + 4 \cdot 23 + 2 \cdot 15 + 2 \cdot 7 + 8 \cdot 30 = 608.$$

The guard residual has degree 44. Their unit-product equation therefore has degree $608 + 44 = 652$. The delivered degree proof records its nonzero leading homogeneous part, and the independent polynomial replay checks the actual specialized circuit. The graph uses 230 operations.

The 748-, 700-, and 664-degree companions retain successively fewer centering operations. They provide explicit control points for both the algebra and the degree/cost tradeoff, rather than unsupported extrapolations.

## 7.4 Fifteen integers and degree 214

The archive also supplies a shorter three-root equation that keeps $f$ as an independent coordinate and the ordinary strong row $I^2 - \Delta(f^2 - 1) = 0$, where $I = (2r+1)c^2$. Here $J = 3j + 1$, $Q = I^2$, and $V = fcT - f^2R - c$; all other

native and guard abbreviations agree with the parent. Set

$$p = 2D_0 + H\{\xi + 3(2\ell + k) + 18V\},$$
$$A = 36H^2A_\tau, \quad D = 4D_m, \quad B_* = 324H^2(QV^2 - 1), \quad \delta = Q - 1,$$
$$U = \delta(p^2 - A - D) - B_*, \qquad L = U^2 - 4\delta^2AD - 4\delta(A + D)B_*,$$
$$N_{15} = L^2 - 64\delta^3ADB_*p^2.$$

The single equation is

$$(1 + 4\{I^2 - \Delta(f^2 - 1)\})(1 + 4G_0)(1 + 4N_{15}) - 1 = 0.$$

Its factors force the strong row, the guard and the three-root norm to vanish. The guard supplies $Q > 1$ before the norm is interpreted; the strong row supplies $3 \mid I$. Fixed valuations then recover the first, auxiliary and main roots. Since $H$ is odd, the doubled carrier restores the main congruence exactly. This is the independently reviewed initial reduction, with no strong normalization or pentagonal substitution. The norm, strong row and guard have degrees 156, 14 and 44, so the final degree is 214. The complete derivation and literal 139-operation circuit are retained in `forms/COMPACT15_PROOF.md` and its companion packet.

# 8 Coefficients, verification, and logical scope

## 8.1 The six fixed integers

The archive's `coefficients/BINDING.json` is authoritative for names, byte encoding and hashes. Integer bytes are minimal unsigned little-endian representations; their compressed wrappers are not part of the mathematical identity.

| Port | Bits | Raw bytes | Definition |
|---|---|---|---|
| $\beta$ | 1,064,921 | 133,116 | $2^{1064921} - 1$ |
| $K_0$ | 1,065,043 | 133,131 | source34 native constructor |
| $M_C$ | 1,064,921 | 133,116 | source34 native constructor |
| $M_F$ | 1,064,922 | 133,116 | source34, already shifted |
| $A_0$ | 1,114 | 140 | $2^{1113}$ |
| $B_0$ | 666,105 | 83,264 | $2^{666104}$ |
| Total | | 615,883 | six fixed numeral payloads |

The native recipe uses a literal 430-action binary machine table and its 943 local tiles. It constructs finite coefficient supports at radix $\rho = 2^{121}$, block length 8801, and outer radix $B = \rho^{8801} = 2^{1064921}$. In its notation,

$$K_0 = D_C + B\rho, \qquad \beta = B - 1, \qquad M_F = M_{F,\text{native}} + \beta.$$

The complete supports, padding, masks and action bytes are supplied, so the recipe is finite and executable without simulating the machine. The final verifier reconstructs all six integers and compares the raw values to their payloads. Source31 and source34 share the same radix exponent but different remaining ports; compact44 is a third tuple. No port substitution between them is made here.

The new equations use the same six large numeral leaves as the calibrated determinant parent, together with small integer multipliers. This does not imply that the fully expanded coefficients have the same maximum height. We do not claim a measured improvement in expanded monomial count or maximum coefficient size. The polynomial is delivered as an exact finite arithmetic graph, not as an enormous expanded list.

## 8.2 What the executable checks establish

The supplied standard-library verifier checks:

1. all actual fixed coefficient bytes and their deterministic reconstruction;

2. literal graph topology, permitted operations, unknown lists, live-node counts, and agreement with the finite construction recipes;
3. the generic denominator-cleared norm templates against a separately written exact multiquadratic-ring evaluator, including signed and rational radicands and explicit roots;
4. exact specialized polynomial degrees and nonzero leading coefficients, with the universal upper bounds justified in the written proof;
5. the normalized Pell, positivity and integral restoration identities, plus explicitly bounded diagnostic cases;
6. integrity of the packaged source files and negative controls showing that coefficient or graph corruption is detected.

These checks do not search for a zero of the consistency polynomial and do not test whether ZFC is consistent. Finite computations do not prove the unbounded native encoding or the universal arithmetic lemmas. The latter are supplied as written proofs and have separate internal mathematical reviews.

### 8.3 Proof status and dependency chain

The logical composition is

$$\text{new integer zero} \longleftrightarrow \text{calibrated source34 zero} \longleftrightarrow \text{finite K215 accepting run} \longleftrightarrow \text{finite ZFC refutation.}$$

The first equivalence is proved here, with explicit signed soundness and canonical completeness. The later two are the named inherited ordinary native/compiler and literal-machine calibration theorems. Their exact source identities are recorded in `SOURCE_PINS.json`; the immediate parent packet and its source archive are included.

All new arguments use finite field extensions, integer valuations, Pell recurrences and classical integer representation theorems, and can be formalized in ZFC. Thus the inherited ordinary calibration composes to the ordinary theorem (1). This report does not claim that the optimized endpoint has been compiled in Lean or Rocq, or that an explicit derivation has been exported in ProveIt's encoded object-language ZFC calculus. The latter distinction is real and is preserved in the retained formalization-status documents. The review here is AI-assisted internal review, not external peer review.

### 8.4 Reproduction

From the extracted archive root run

```
python3 verify.py
python3 -O verify.py
```

Both modes use the Python standard library. No network, machine-halting search, numerical square-root approximation or proof assistant is required for this finite replay. Output reports state their exact scope. To rebuild the article, from `report/` run twice:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The release PDF was compiled and visually inspected separately from the arithmetic replay. The ZIP contains complete LaTeX sources, the PDF, literal equations, numerical coefficients, construction and verification code, proof notes, independent reviews and provenance.

## 9 What this changes, and what remains open

The practical contribution is a new modest-degree range for equations over *unrestricted integers*. The inspected small-degree chain previously ended at seventeen variables and degree 88; the new points at fourteen and thirteen variables have degrees 652 and 1392. The huge-degree thirteen- integer construction is no longer the only delivered thirteen-integer option in this research line.

The reusable idea is a local arithmetic invariant: make every relevant radicand a unit at one prime, then assign distinct valuations through fixed weights. It removes the need to separate every possible signed subset by large real inequalities. The strong normalization and positivity conversion are useful precisely because they preserve this invariant. Applying the combiner without those uniform residue checks would be unsound.

The contribution does not lower the ten-integer or nine-positive general variable counts, solve the consistency question, establish worldwide priority, or prove that any delivered equation is globally simplest. It also retains substantial inherited coefficient data. A further improvement should measure variables, degree, live circuit size and coefficient size separately, and should preserve the explicit endpoint calibration.

The next concrete formalization target is now small and well delimited: the valuation-separated square-class lemma, the normalized odd-index Pell adapter, the 3-adic-unit positivity lemma, and the exact five-root norm identity. None requires redoing the native machine compiler. Completing those kernel proofs would strengthen verification of the new reduction; the native source-to-refutation internalization remains its own task.

# A The immediate seventeen-integer parent

The literal parent is retained as `upstream/transport/degree88_17.json`. It uses the same native and guard abbreviations, with $J = 3J_3 + 1$, ordinary strong variable $f$, and three separate root coordinates. Put

$$I = (2r+1)c^2, \quad V = fcT - f^2R - c, \quad D = X + ac + H\gamma, \quad P = \ell + k.$$

Its five residuals are

$$\begin{aligned}
&\tau^2 + 2\tau(2P - k) + k^2 - 4,\\
&D^2 - \Delta c^2 - 1,\\
&I^2 - \Delta(f^2 - 1),\\
&V^2 + (1 - I^2)y(2V + y) - 1,\\
&GZ - (s^2 - G - Z + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3).
\end{aligned}$$

The parent equation is their sum of squares. Its unknowns are $J_3, C, z, f, h, r, v, w, \tau, y, \gamma, T, c, s, p_1, p_2, p_3$. The definitions in Sections 5–6 give the exact maps needed to compare this parent with the new equations.

# B The source boundary is a mathematical interface

The retained parent uses the calibrated determinant condition $G > 0$ and the interval condition $\eta\zeta > 0$. Soundness of the determinant applies to arbitrary signed assignments; completeness uses fixed leading-digit bounds of canonical native witnesses. The residue orientation imposes $X \equiv 2 \pmod 3$ to select the positive signed power of two. The CRT compression of auxiliary quotients uses the strong rank lemma before the auxiliary rank is identified, avoiding a circular use of the target rank. The original proofs of these facts are retained with the parent archive.

The present reduction does not strengthen any of these inherited interfaces silently. The only additional canonical restrictions are $J \equiv 1 \pmod 6$ and the normalized odd strong witness. The first follows from the already stated canonical repunit; the second is constructed in Section 6. The pentagonal representation holds for every positive guard value. All compatibility claims therefore have explicit proofs.

# References

[1] V. Reshetnikov and contributors, *ProveIt*, comparison commit `cfbf118b4ad9b2c651fe96b571d9415261766328`, the DiophantineConsistency collection, active formalization documents and incoming archives. https://github.com/VladimirReshetnikov/ProveIt/tree/cfbf118b4ad9b2c651fe96b571d9415261766328/Computability/DiophantineConsistency.

[2] *Pell congruences and Diophantine consistency*, incoming archive `con_zfc_pell_congruences_2026-10-23.zip`, including the degree-88 seventeen-integer CRT/transport parent. Original archive, proofs, coefficient constructor and source notices retained in this release.

[3] ProveIt packages 34, 61, 69, 70 and 71: source34 mixed-parity native coefficients; signed auxiliary-index and squared-modulus reconstruction; rational Pell descent; residue orientation; calibrated determinant guard. All referenced source identities are pinned in the release provenance.

[4] Z.-W. Sun, *On universal sums of polygonal numbers*, Science China Mathematics **58** (2015), 1367–1396. https://arxiv.org/pdf/0905.0635. In particular the discussion of Guy's three-generalized-pentagonal observation and the norm-9 identity (1.4).

[5] B.-K. Oh, *Ternary universal sums of generalized pentagonal numbers*, Journal of the Korean Mathematical Society **48** (2011), 837–847. https://arxiv.org/pdf/0911.1181.

[6] Y. Matiyasevich and J. Robinson, *Reduction of an arbitrary Diophantine equation to one in 13 unknowns*, Acta Arithmetica **27** (1975), 521–553. https://doi.org/10.4064/aa-27-1-521-553. Classical background for relation combining; the fixed-prime specialization used here is proved directly in Section 3.